

Algorithms



Lecture 8.1 Graph Basics

Prantik Paul [PNP]

Lecturer

Department of Computer Science and Engineering

BRAC University

Adapted From: Slides by Yosuke Mizutani

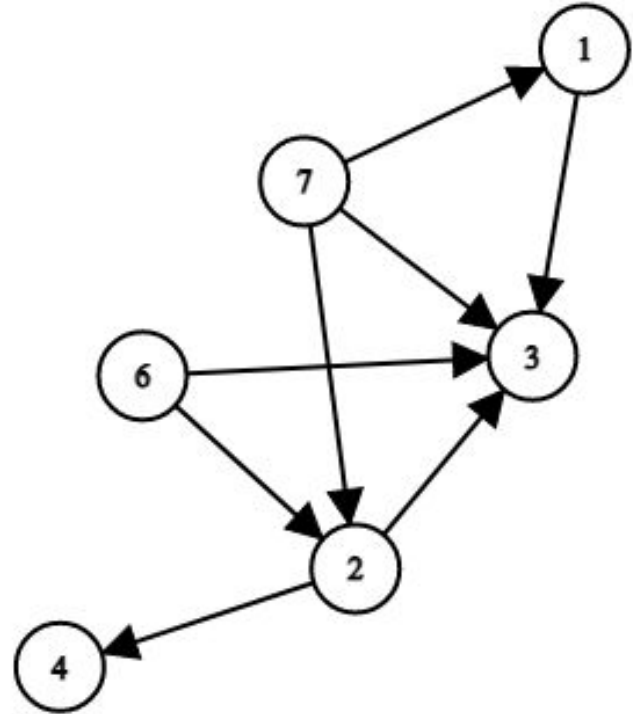
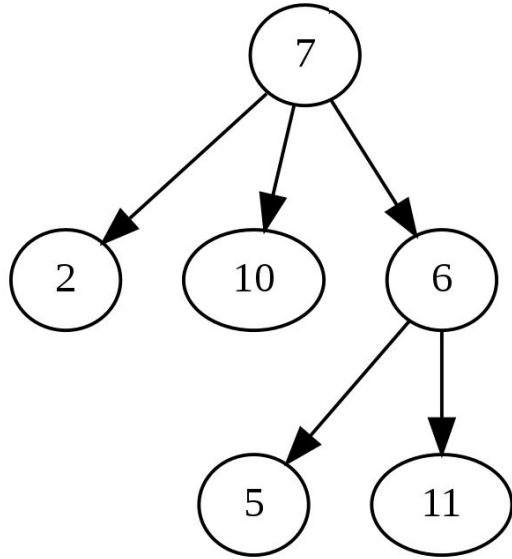
Graph - A superset of Trees

→ $G := (V, E)$

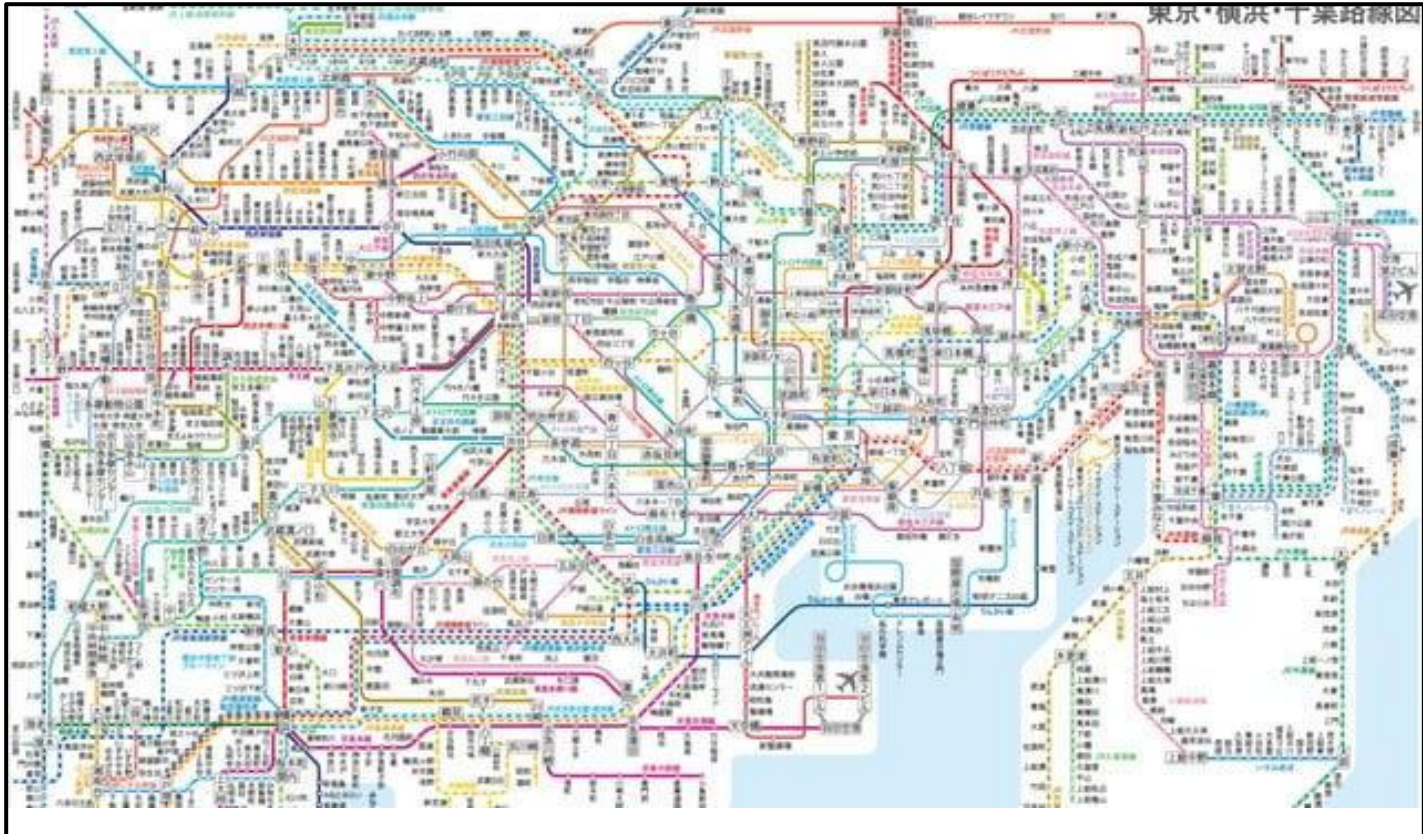
→ $V :=$ Set of vertices (nodes)

→ $E :=$ Set of edges

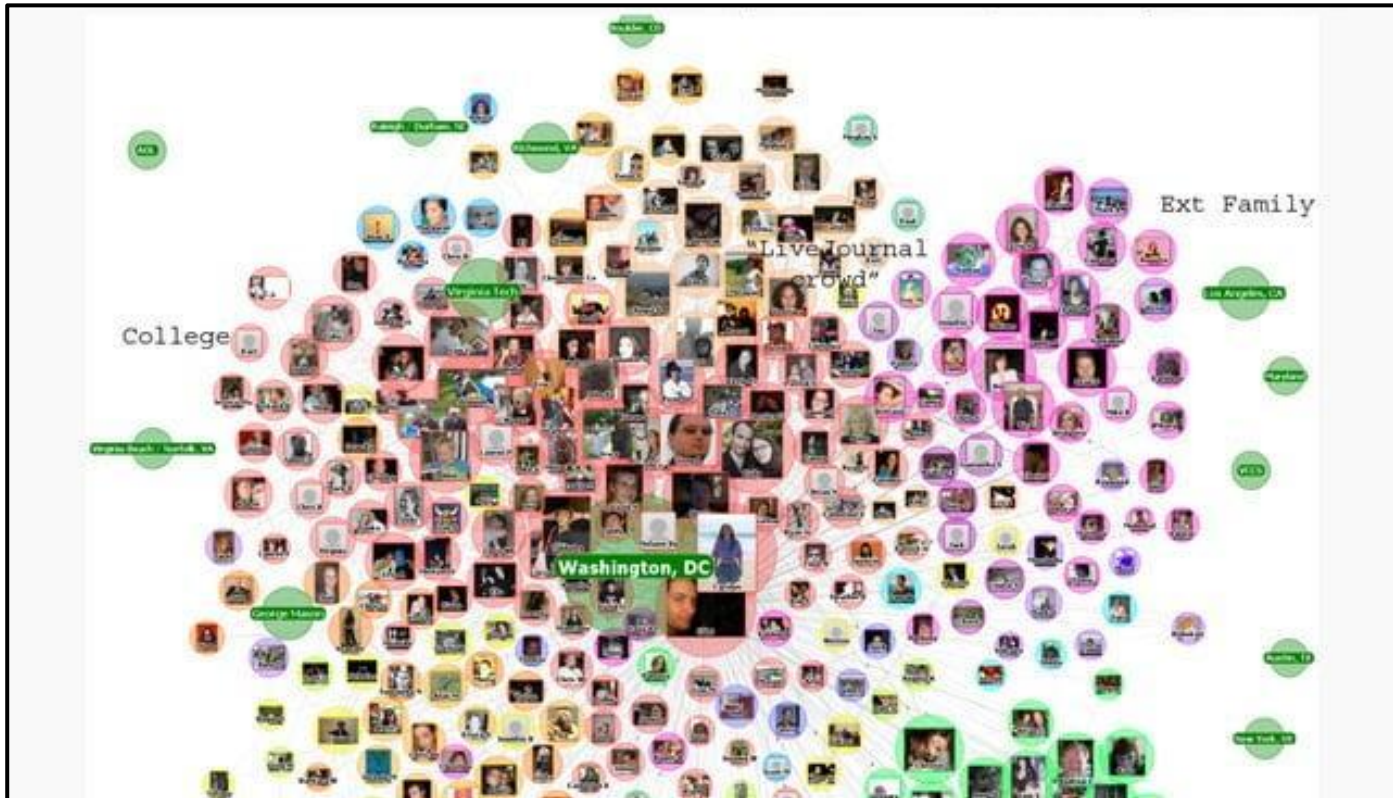
Graph - A superset of Trees



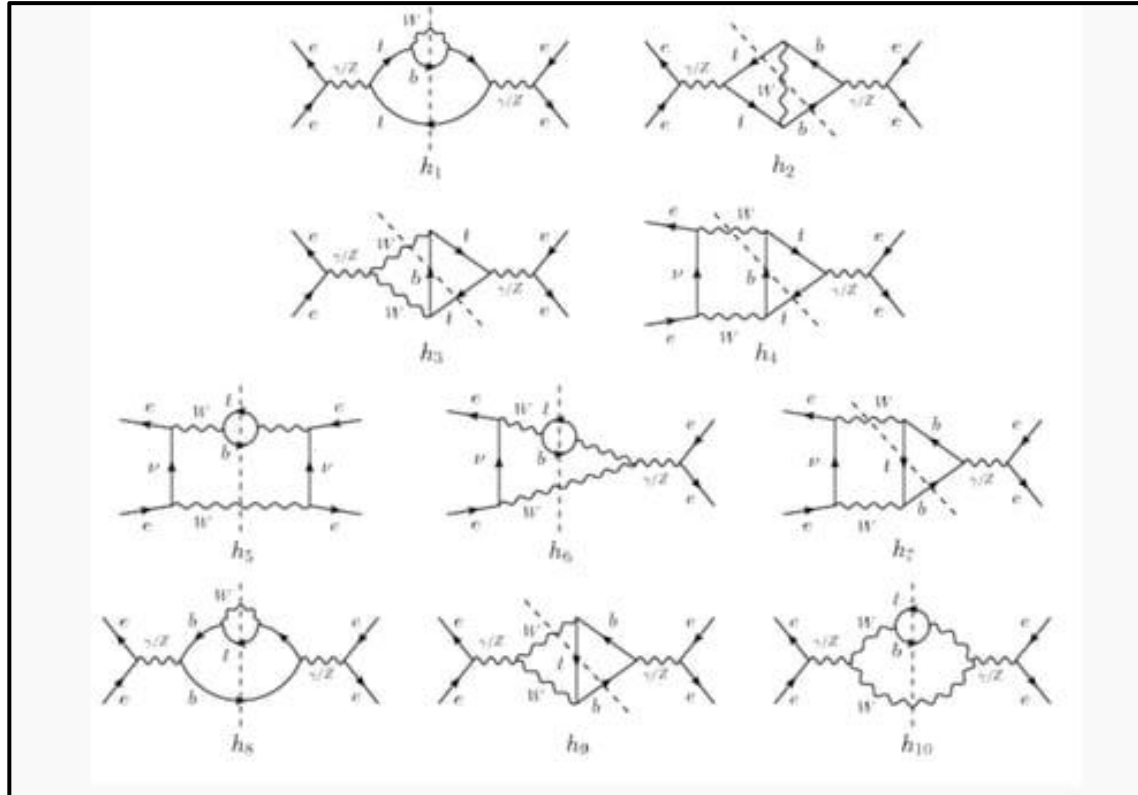
Graphs - IRL (Rail Map)



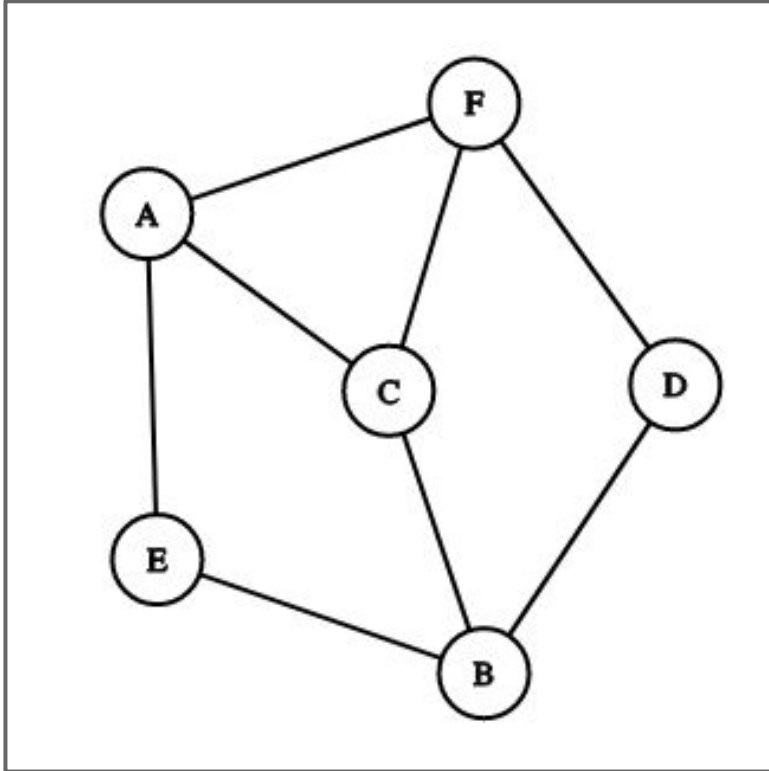
Graphs - IRL (Social Network)



Graphs - IRL (Circuit Diagrams)



Types of Graphs

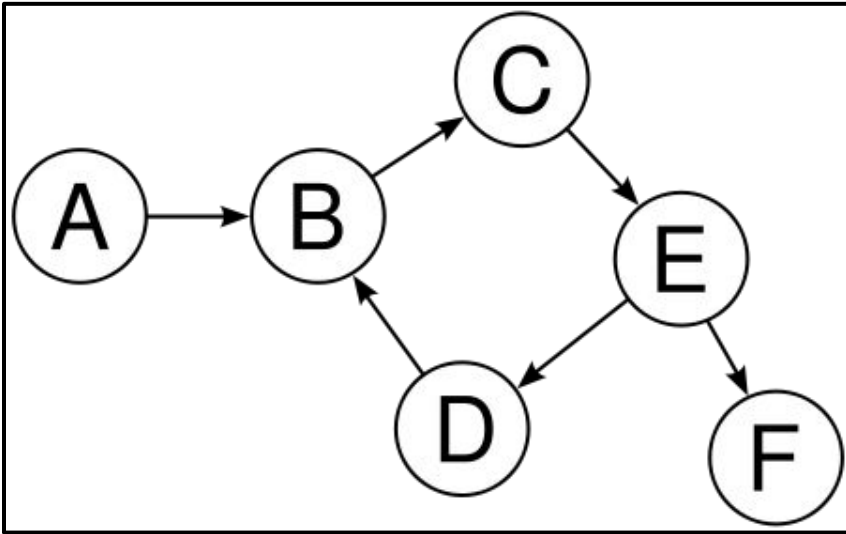


Undirected Graph

$V = \{A, B, C, D, E, F\}$

$E = \{ \{A,E\}, \{A,C\}, \{A,F\}, \{B,C\}, \{B,D\}, \{B,E\}, \{C,F\}, \{C,D\}, \{D,F\} \}$

Types of Graphs



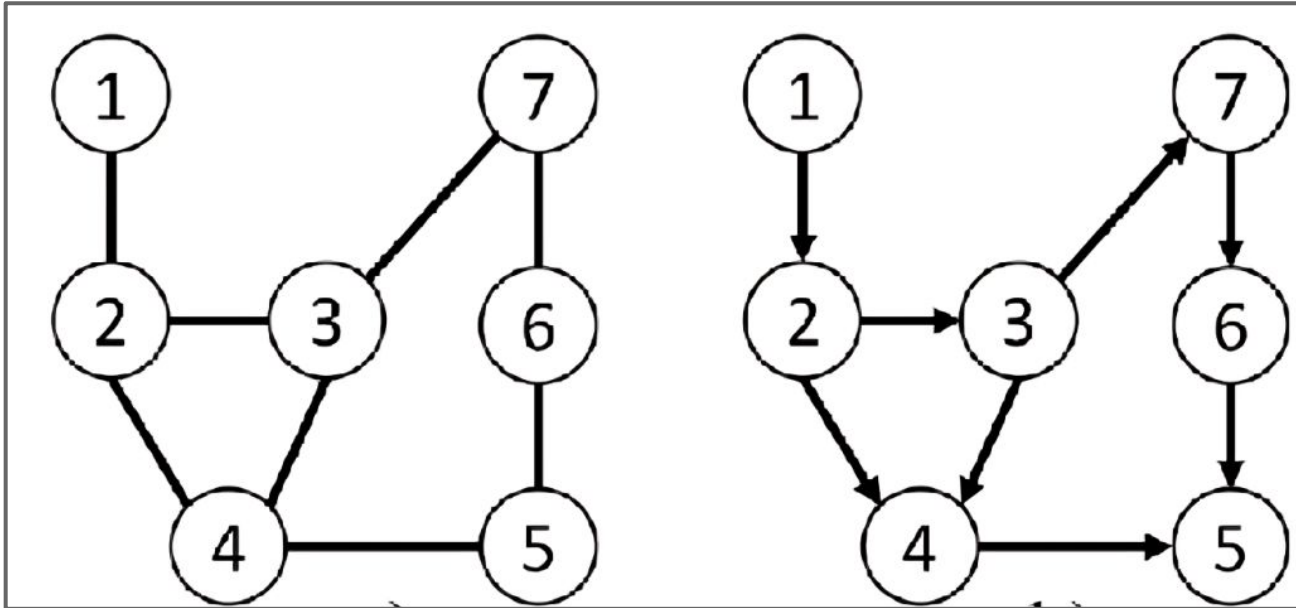
Directed Graph

$V = \{A, B, C, D, E, F\}$

$E = \{ \{A,B\}, \{B,C\}, \{C,E\}, \{D,B\}, \{E,D\}, \{E,F\} \}$

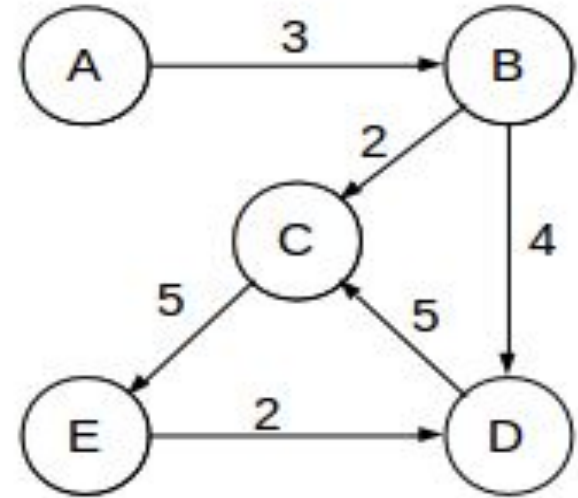
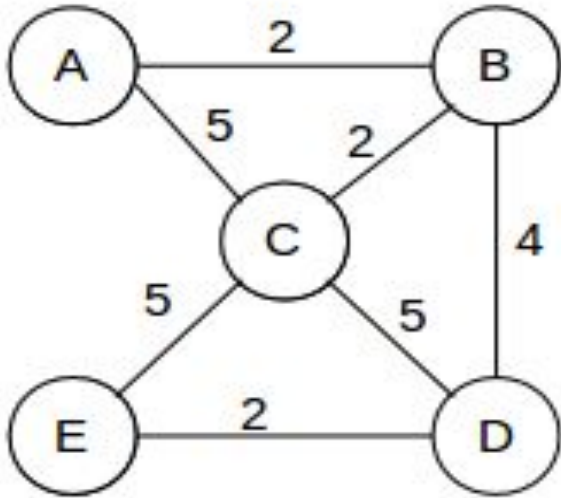
Types of Graphs

Unweighted Graph



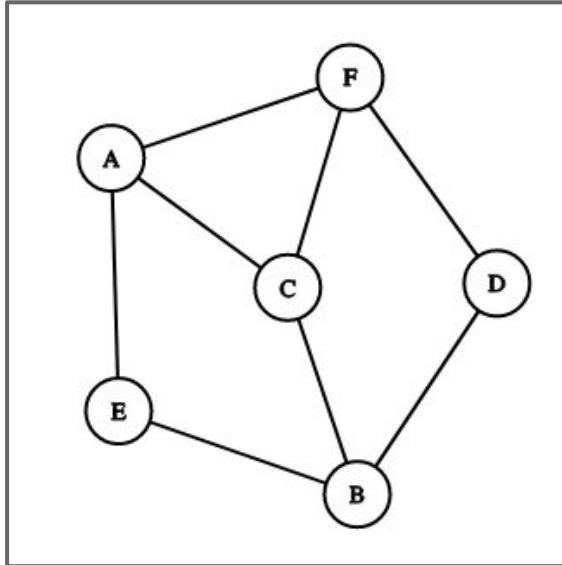
Types of Graphs

Weighted Graph



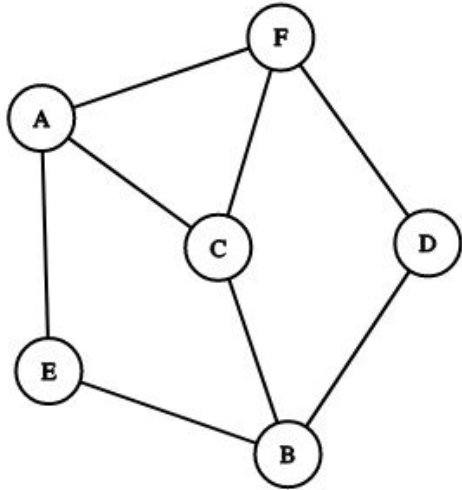
Graph - Basic Terminologies

Edges : Basic unit connecting vertices



Graph - Basic Terminologies

Adjacent Vertices: Vertices sharing an edge

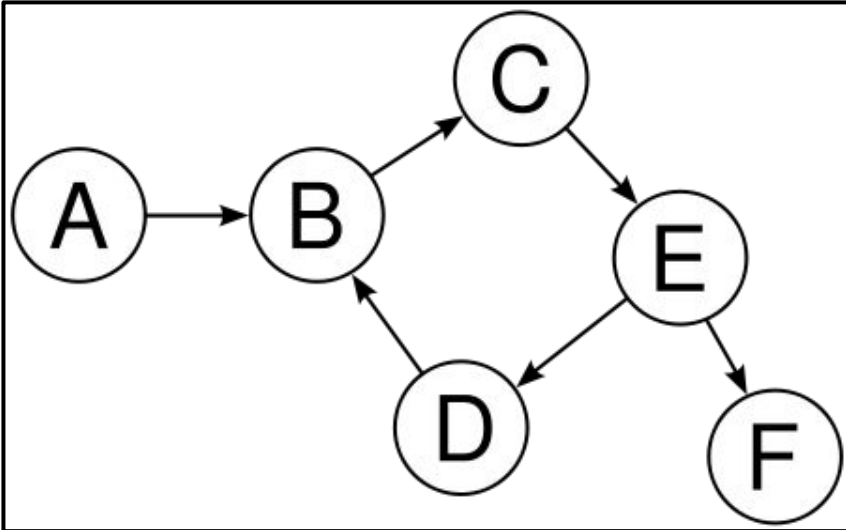


$$\text{Adj}(A) = \{E, C, F\}$$

$$\text{Adj}(C) = \{A, B, F\}$$

Graph - Basic Terminologies

Adjacent Vertices: Vertices sharing an edge



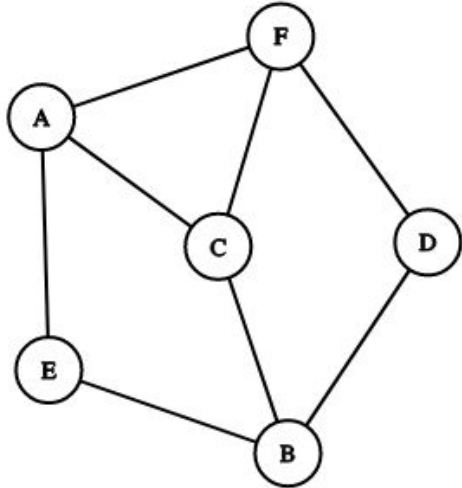
$$\text{Adj}(A) = \{B\}$$

$$\text{Adj}(B) = \{C\}$$

$$\text{Adj}(E) = \{D, F\}$$

Graph - Basic Terminologies

Degree : Total number of edges connected to a vertex



$$\deg(A) = 3$$

$$\deg(D) = 2$$

$$\deg(B) = 3$$

$$\deg(E) = 2$$

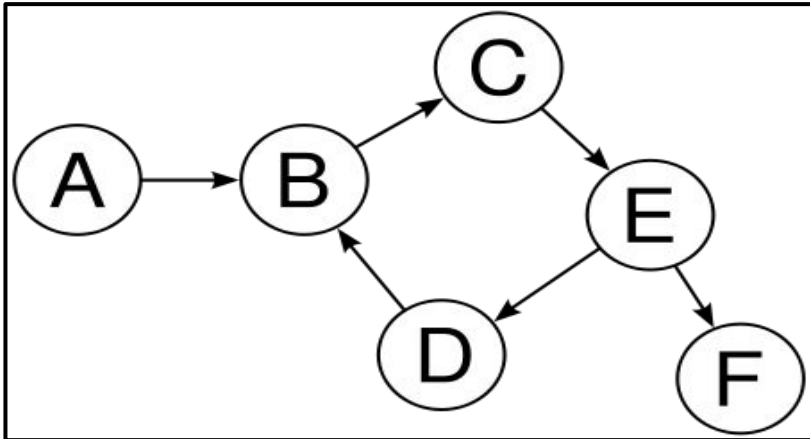
$$\deg(C) = 3$$

$$\deg(F) = 3$$

Graph - Basic Terminologies

Incoming Edges: directed edges reaching a vertex

In Degree : total number of incoming edges to a vertex



$$\text{Indeg}(A) = 0$$

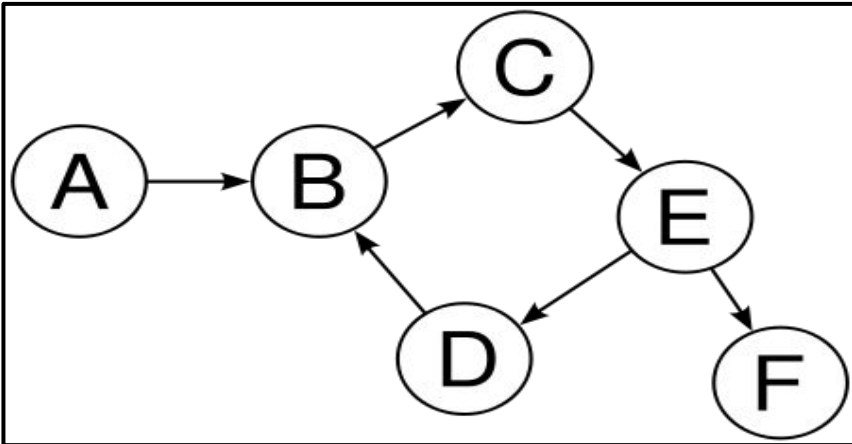
$$\text{Indeg}(B) = 2$$

$$\text{Indeg}(C) = 1$$

Graph - Basic Terminologies

Outgoing Edges: directed edges starting from a vertex

Out Degree: total number of outgoing edges from a vertex



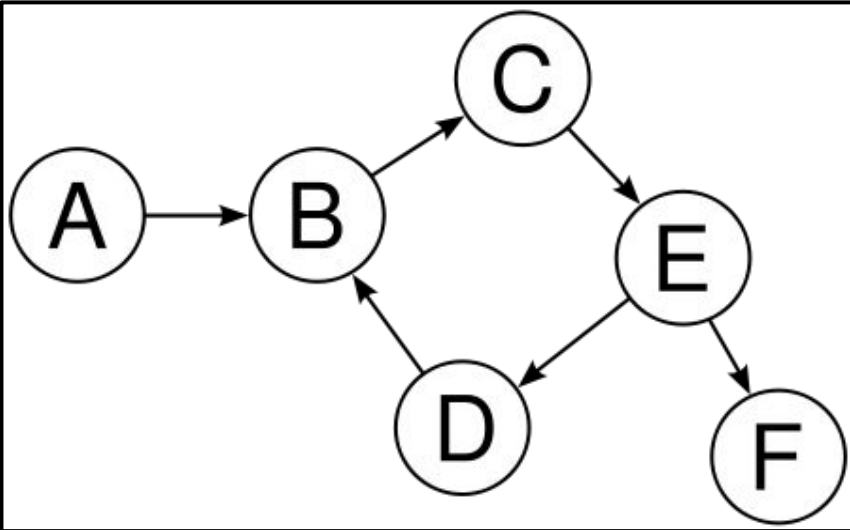
Outdeg(A) = 1

Outdeg(E) = 2

Outdeg(F) = 0

Graph - Basic Terminologies

Degree: Indegree + Outdegree



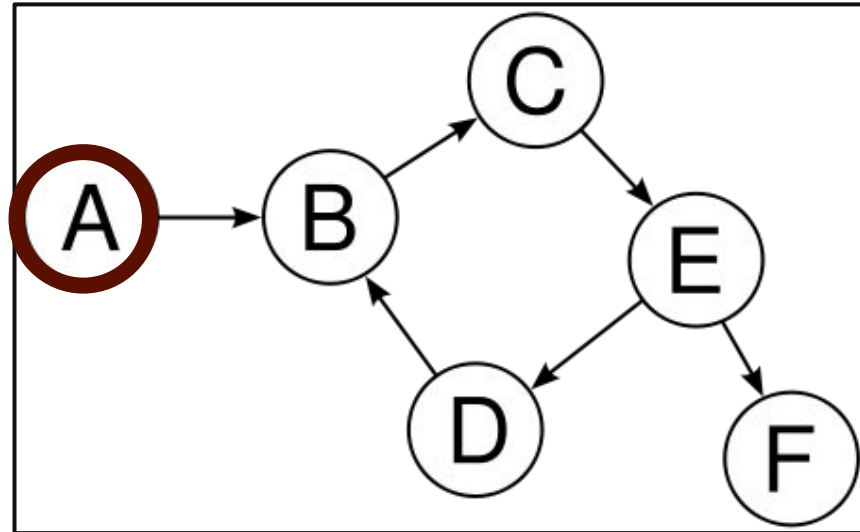
$$\deg(A) = 1$$

$$\deg(E) = 3$$

$$\deg(F) = 1$$

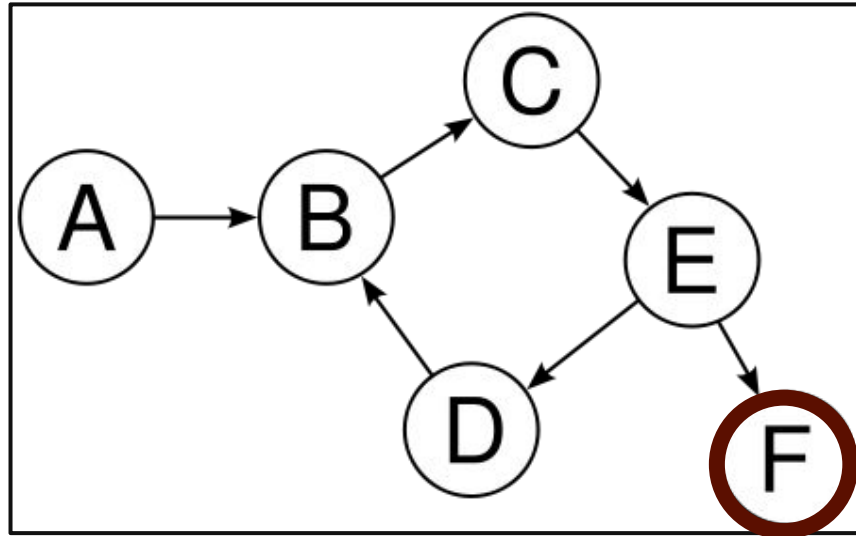
Graph - Basic Terminologies

Source Vertex : A vertex with in-degree zero



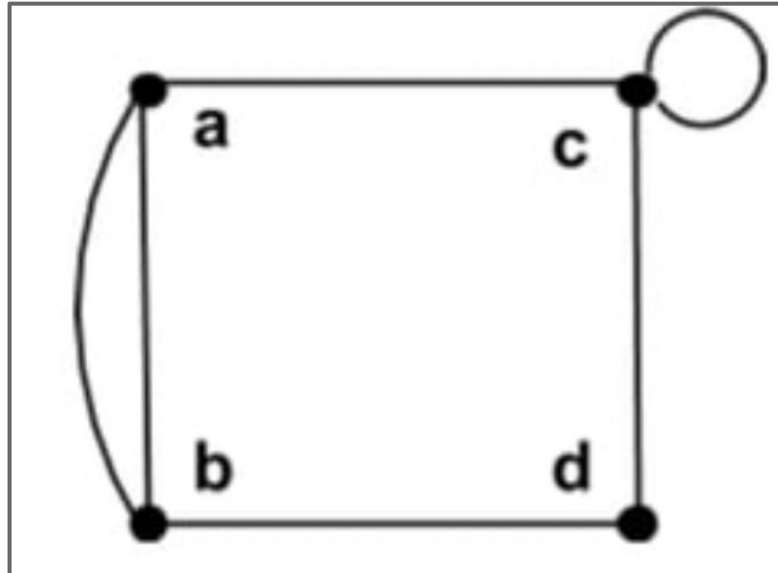
Graph - Basic Terminologies

Sink Vertex : A vertex with out-degree zero



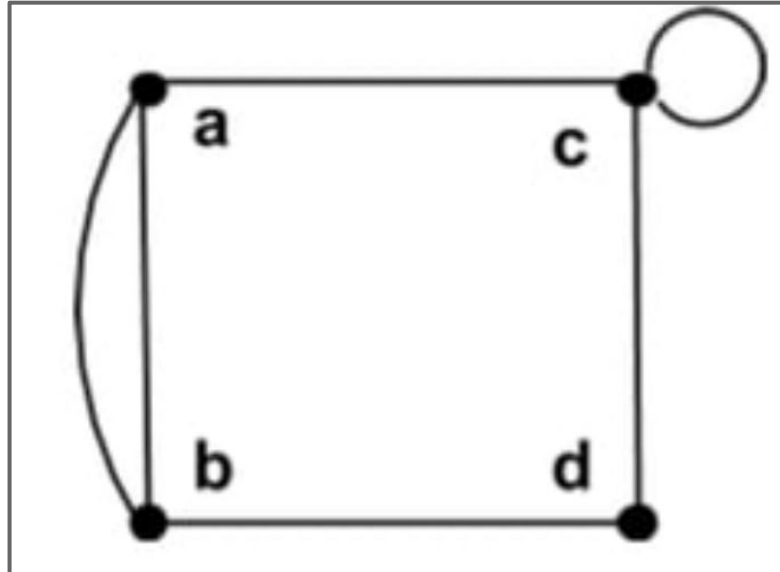
Graph - Basic Terminologies

Parallel Edges : Multiple edges between the same pair of vertices



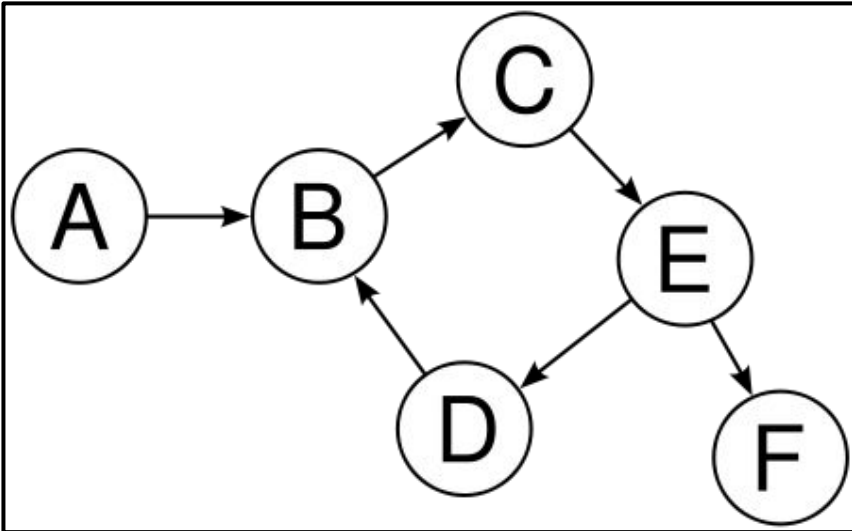
Graph - Basic Terminologies

Self Loop : Edge between a vertex and itself



Graph - Basic Terminologies

Path : sequence of vertices where each adjacent pair is connected by an edge



Path(A, F)

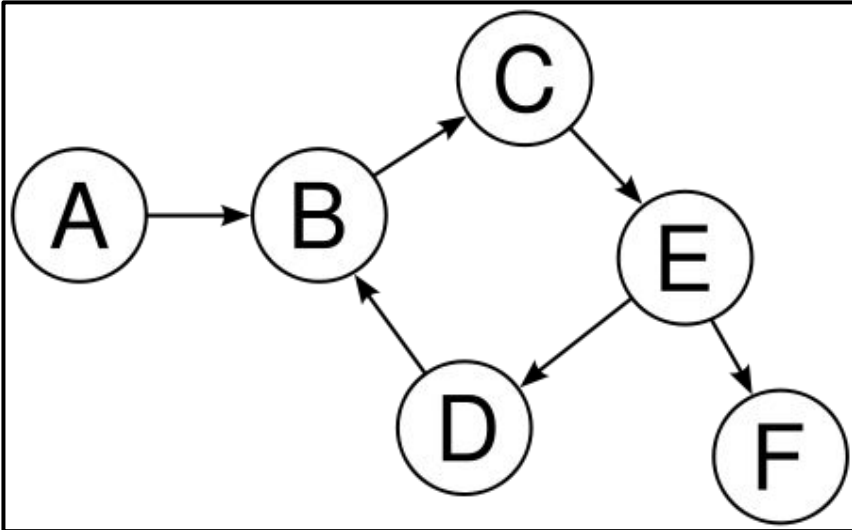
A → B → C → E → F

Path(D, A)

No path

Graph - Basic Terminologies

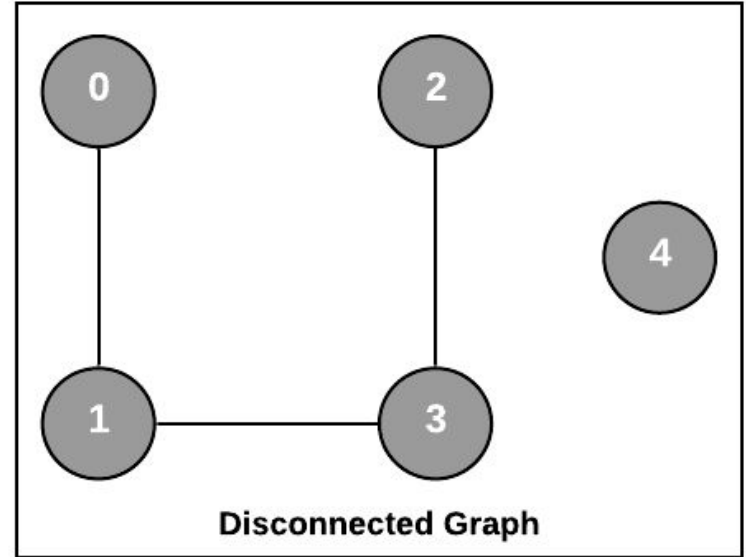
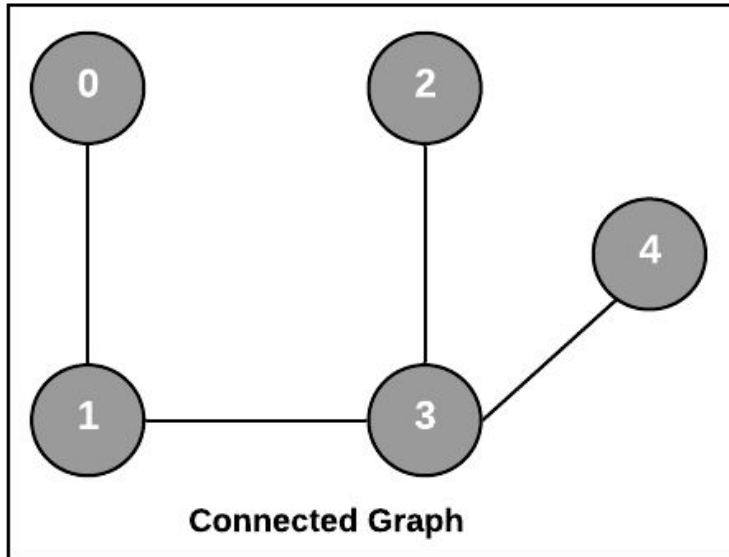
Cycle : closed path where the first and last vertices are the same



B → C → E → D → B

Graph - Basic Terminologies

Connected Graph: Has a path between every pair of vertices



Graph - Basic Terminologies

Connected Graph: Has a path between every pair of vertices

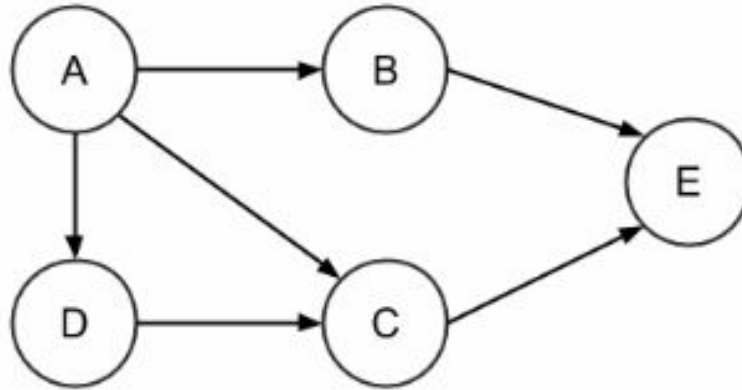
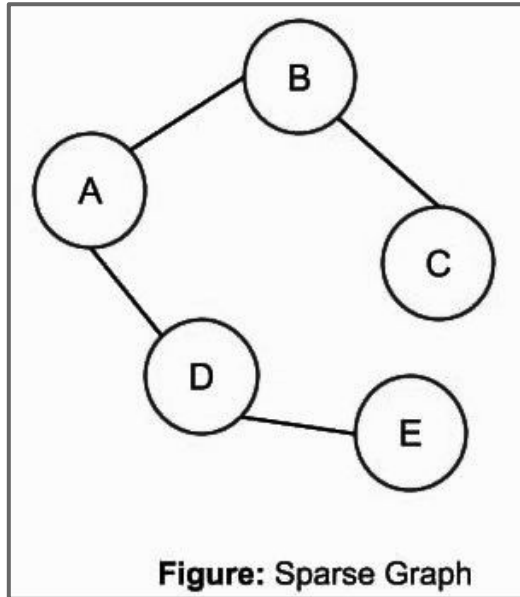


Figure: Disconnected Graph

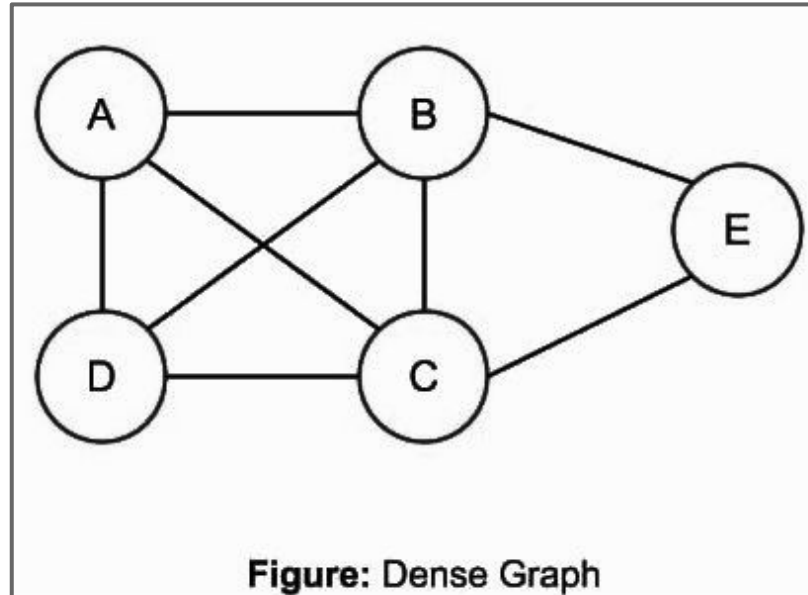
Graph - Basic Terminologies

Sparse Graph: number of edges is considerably less than the maximum number of edges

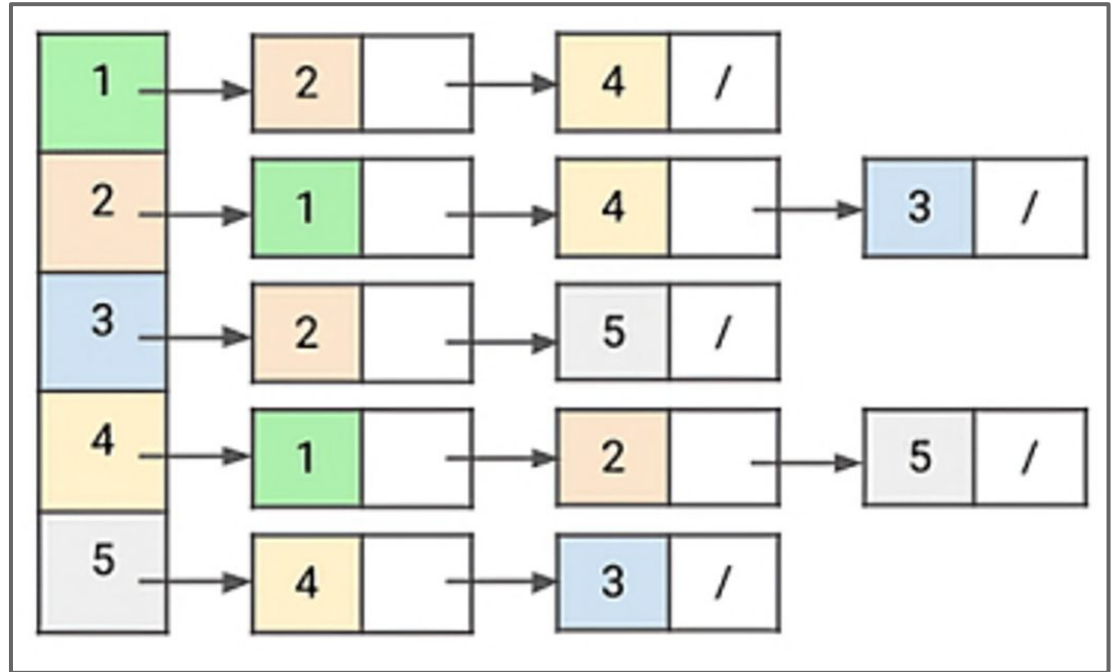
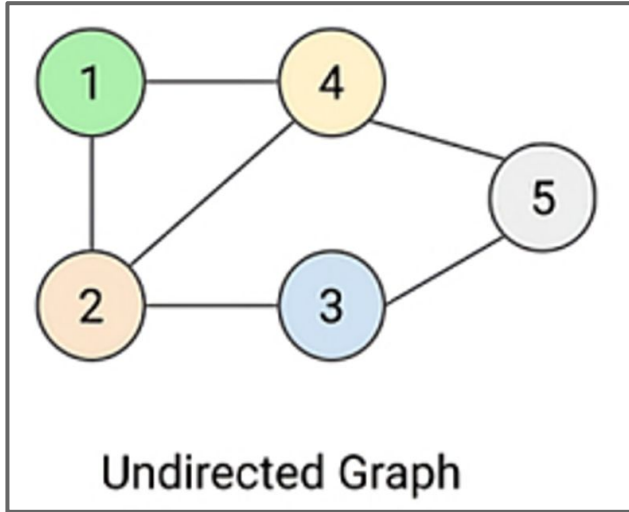


Graph - Basic Terminologies

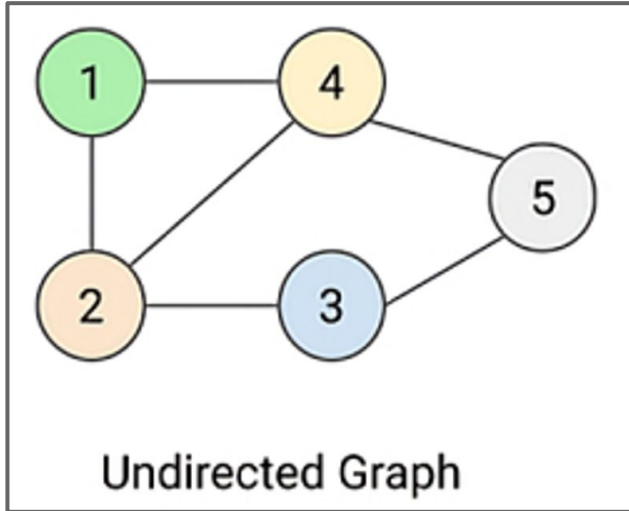
Dense Graph: number of edges is close to the maximum number of edges



Graph - Representation - Adjacency List

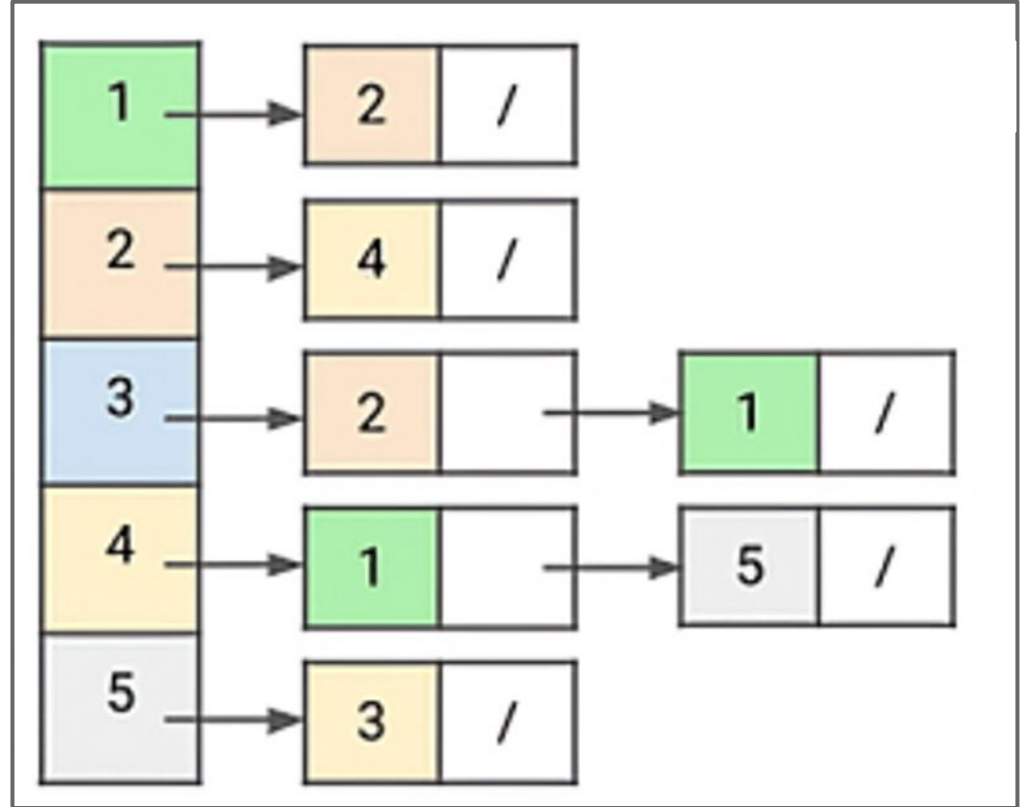
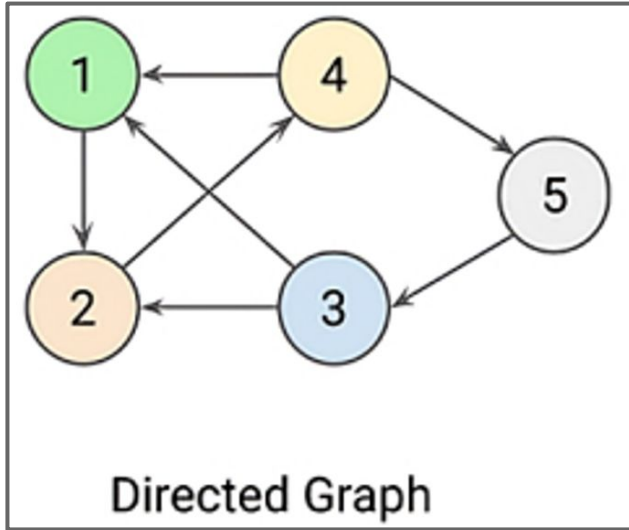


Graph - Representation - Adjacency Matrix

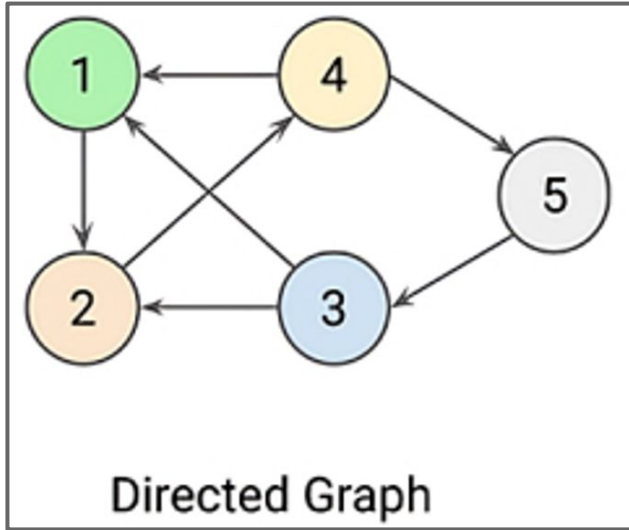


	1	2	3	4	5
1	0	1	0	1	0
2	1	0	1	1	0
3	0	1	0	0	1
4	1	1	0	0	1
5	0	0	1	1	0

Graph - Representation - Adjacency List



Graph - Representation - Adjacency Matrix



	1	2	3	4	5
1	0	1	0	0	0
2	0	0	0	1	0
3	1	1	0	0	0
4	1	0	0	0	1
5	0	0	1	0	0

Graph - Representation

What to do for Weighted Graph's Adjacency List Representation?

**Store Tuples like <Vertice, Weight>
Instead of only Vertice**

Graph - Representation

What to do for Weighted Graph's Adjacency Matrix Representation?

**Store Weight in the Matrix Entry
Instead of 0 or 1**